

A SCIENCE & ENGINEERING TALK

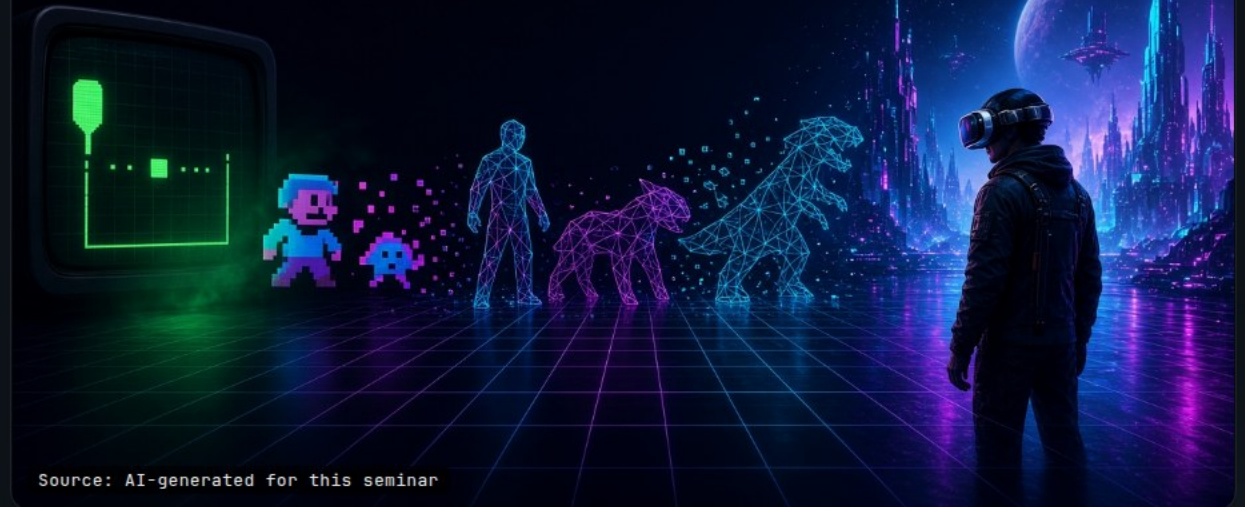
The Engineering of Play

How games grew up — and where they are going — told through the engineering that made each leap possible.

Soutam Rajbhar

Department of Chemical Engineering
IIT Kharagpur

Pong on a TV → 8-bit characters → 3D → stepping into a virtual world



Source: AI-generated for this seminar

WHY THIS MATTERS

Gaming is the **biggest** entertainment business on Earth.

Bigger than movies and music put together. And it isn't really an art industry. It's an engineering one.

~\$200B

MONEY MADE EACH YEAR

3.3B+

PEOPLE WHO PLAY

1 in 3

PEOPLE ON THE PLANET PLAY

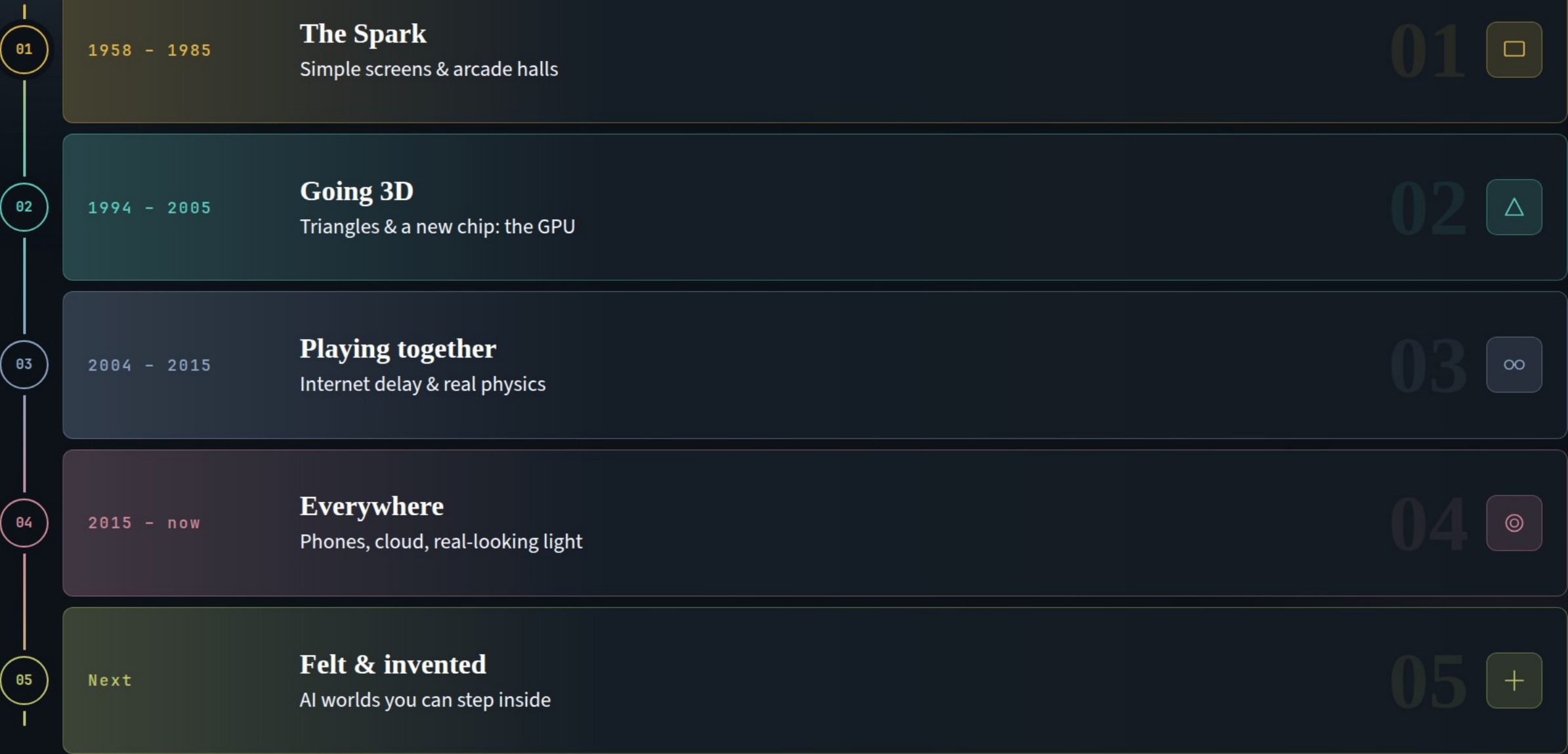
60–240

NEW PICTURES DRAWN EVERY
SECOND

Every second, your computer does maths for millions of points — shape, bounce, light — and paints a new picture before you can blink. That's the story we're following today.

Five big leaps. One story.

Each leap happened because engineers solved a hard problem. Here's the map for today.



The Spark: making a screen talk back

Before fancy pictures, the hard question was simple: can a machine react to a person right now?



The games

Tennis for Two (1958) on a lab screen, *Pong* (1972), then *Space Invaders* and packed arcade halls.

How it was built

Early machines were wired by hand. There was often no “brain chip” yet — the wiring *itself* was the game.

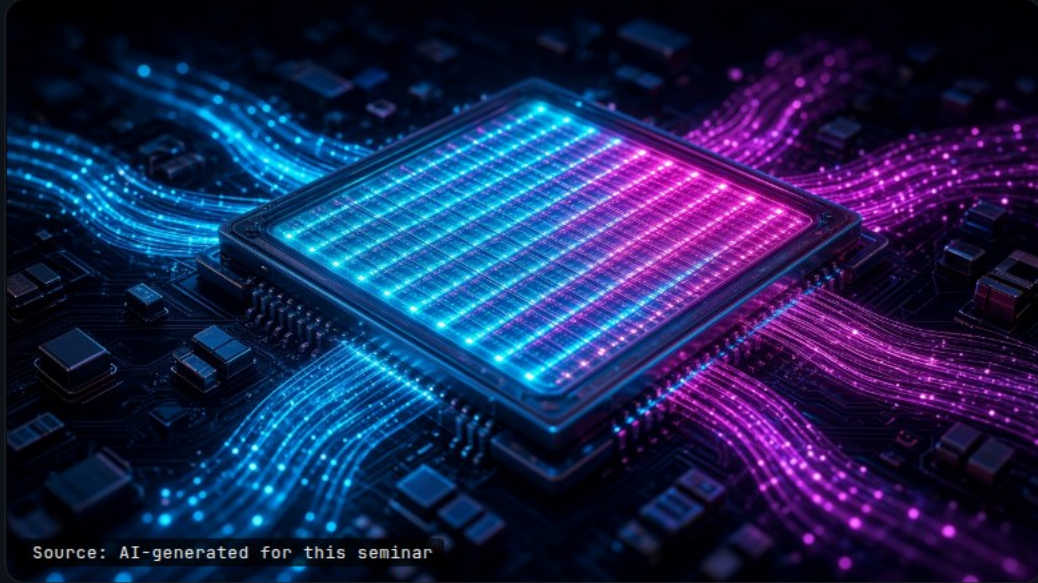
The big change

Cheap computer chips arrived. A game became *software* — the same machine could play many different games.

Takeaway: once games were software, progress stopped being about new wires and started being about faster computers.

Games go 3D — and a new chip is born: the GPU

To turn flat pictures into worlds, a computer had to do the same maths millions of times — very fast.



The problem

A 3D world is made of millions of tiny triangles. Every picture, the computer must move them, light them, and colour every dot. A normal chip (the CPU) does this one step at a time — too slow for smooth play.

The answer

A new chip built just for pictures: the **Graphics Processing Unit**. It has many small workers that colour lots of dots *at the same time*. NVIDIA's 1999 GeForce 256 was sold as the first of these.

PlayStation

Nintendo 64

Quake

GeForce 256

One idea matters more than anything else today: do the same small job on thousands of things at once. Let's watch it happen. →

Why a GPU? Many hands at once.

Both chips colour the same 144-dot picture. The CPU has 4 strong workers. The GPU has 48 smaller ones. Watch the clock.



shaded

0/144

time

0.00s

CPU · 4 cores

GPU · 48 cores

CPU · 4 cores
Few workers. Each one is smart and can do many kinds of jobs — like 4 master chefs in a kitchen.

GPU · 48 cores
Many workers. Each one is simple and repeats one job — like 48 line cooks all chopping vegetables at once.

If we gave the CPU 48 cores, would it become a GPU? No. More chefs still aren't a factory line. A real GPU has thousands of these simple cores. That same trick now trains AI.

Playing together: **delay**, physics & huge worlds

Home internet turned games into places we share. That created new headaches for engineers.

01 Fighting delay

A signal can take about 40 milliseconds to cross the world. Games guess your next move, then correct it — so play still feels instant.

01

02 Real physics

Games solve bounce, crash, and falling bodies 60 times a second — the same kind of maths used in engineering simulations.

02

03 Huge worlds

Games like *World of Warcraft* keep tens of thousands of players in sync across many computers at once.

03

The shift: a game stopped being “a program on your computer” and became a live system serving millions of people at once.

Games go everywhere

From a phone in every pocket to light that looks almost real.

01



In every pocket

Most gaming now happens on phones. A modern phone already has a strong picture chip — a console in your hand.

01

02



Games in the cloud

The game can run on a faraway computer and send you video — like Netflix, except your button press has to travel there and back in a blink.

02

03



Tracing light

New chips follow rays of light through a scene — real reflections and shadows. Movie computers used to spend hours on one frame. Games now do it live.

03

04



Esports

Competitive games fill stadiums. That needs rock-solid servers, fair play, and timing accurate to a fraction of a picture.

04

Gaming tech quietly **rebuilt the world**

Tools made so we could have fun ended up powering some of today's most serious work.

01 GPUs → the AI boom

The same “many hands” maths that colours pixels is what trains AI. Chatbots and image models run on chips that started life in games.

01

02 Game tools → film & design

The same software that builds games now draws movie sets, car showrooms, and buildings — live, not overnight.

02

03 Game physics → factories

Robots and self-driving cars practise in game-like worlds. Factories test a virtual copy of a plant before they build the real one.

03

The point: chasing a fun, hard problem — drawing a world 60 times a second — gave every other field a new set of tools.

Worlds that are **invented**, not just drawn

The next leap: the computer doesn't only paint the world. It starts to imagine it.

01 AI fills in the picture

Games already draw a rough picture, then let AI add the extra detail — more sharpness for less work. NVIDIA calls one version of this DLSS.

01

02 Worlds made on the spot

Research models can now invent playable game pictures in real time. Imagine levels, characters, and quests made just for you.

02

03 Characters that talk back

Chat-style AI can give game characters real conversations — they reply to what you actually say, not a fixed script.

03

04 Endless maps + AI

Older tricks already built whole galaxies from one seed number. Pair that with AI and you get huge worlds that still feel surprising.

04

From watching a screen to a world you **step inside**

The next race isn't only prettier pictures. It's making the world feel close enough to touch.



Source: AI-generated for this seminar

Headsets & mixed reality

A headset must draw two sharp pictures, many times a second, with almost no delay after you move your head. Miss that, and people feel sick.

Feeling the world

Motors in controllers and suits let you feel weight, texture, and kickback — not just see it.

Play with almost no hardware

As networks get faster, the heavy work can live in the cloud — a rich world streamed to thin, cheap glasses.

Playing by thought

Early brain-computer links already let some people move a cursor by thinking. That's a far-off idea of how we might one day play.

The same story, over and over

01

A wild wish

Someone wants a rich world that reacts right now.

02

A clever leap

Engineers answer with a new chip or a smart trick — the GPU, delay-hiding, tracing light, AI drawing.

03

It leaves the game

That trick becomes a tool for everyone — AI, film, robots, and science.

Games are where we practise building worlds that react in real time — and that practise keeps handing the rest of engineering its next tools.

Key takeaways

1

Gaming is an engineering story

Bigger than film and music put together — and every leap was won by engineers, not only artists.

2

The GPU is the hero chip

Built to colour dots side by side, it now trains almost every big AI model. Fun hardware became world-changing hardware.

3

Game tricks keep leaving the game

Picture chips, physics, online play, and AI drawing all leaked out to power AI, film, robots, and science.

4

The future is invented and felt

AI will imagine worlds as you play them, and headsets, touch, and networks will make those worlds feel close.

WHERE THE FACTS COME FROM

References

A starting list if you want to read more after today.

01 Newzoo. Global Games Market Report — how much money and how many players · 2024

03 NVIDIA. DLSS — AI that adds extra detail to a game picture · 2020–

05 Valve Developer Community. Source Multiplayer Networking — hiding internet delay

07 Google Research. GameNGen — a neural net that can run a playable game · 2024

02 NVIDIA. GeForce 256 — sold as “the world’s first GPU”, plus later chip history · 1999–

04 Akenine-Möller, Haines & Hoffman. Real-Time Rendering (4th ed.) — how live pictures are drawn · 2018

06 Google DeepMind. AlphaGo & AlphaStar — AI that learned by playing games · 2016–19

08 Cooper et al., Nature. Foldit — a game that helped fold a real protein · 2010

Numbers are rounded for scale. Every picture carries its own source line. Pictures in this talk are original AI-generated illustrations made for this seminar — not photos of real products.

ONE LAST THOUGHT

The next leap in technology won't be announced in a lab — it'll ship in a game.

Dots → triangles → many hands at once → AI → worlds you can step inside. From *Pong* on a lab screen to places we may one day walk into — every leap was an engineering leap.

Watch what gamers play next. It's a preview of what everyone else will build with.

Thank you — questions welcome

Soutam Rajbhar · Chemical Engineering, IIT Kharagpur